HDOC Day-13 Activity

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Count the number of even digits and odd digits.

2. Find the sum of its even digits and the sum of its odd digits separately.

Step 1: Understanding the Problem

Input: A positive integer n and menu choice.

Output: For choice 1, number of even and odd digits. For choice 2, sum of even digits and sum of odd digits.

Method: Extract digits using % 10 and remove digits using integer division by 10. Use switch-case for menu selection.

Step 2: Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	n=12045, choice=1	Even_dig=3, Odd_dig=2	Even_dig=3, Odd_dig=2	Pass
2	n=12345, choice=1	Even_dig=2, Odd_dig=3	Even_dig=2, Odd_dig=3	Pass
3	n=12345, choice=2	Sum_even=6, Sum_odd=9	Sum_even=6, Sum_odd=9	Pass
4	n=2468, choice=2	Sum_even=20, Sum_odd=0	Sum_even=20, Sum_odd=0	Pass
5	n=1357, choice=2	Sum_even=0, Sum_odd=16	Sum_even=0, Sum_odd=16	Pass

Step 3: Algorithm and Evaluation

1. Start.

2. Read a positive integer n.

3. Display menu: 1 - Count even and odd digits; 2 - Sum even and odd digits.

4. Read choice.

5. Use switch(choice).

6. For choice 1:

a. Set even_count = 0 and odd_count = 0.

b. Copy n into temp.

c. While temp > 0, extract digit = temp % 10.

d. If digit % 2 == 0, increment even_count; otherwise increment odd_count.

e. Remove the last digit using temp = temp / 10.

f. Display Even_dig and Odd_dig.

7. For choice 2:

a. Set sum_even = 0 and sum_odd = 0.

b. Copy n into temp.

c. While temp > 0, extract digit = temp % 10.

d. If digit is even, add it to sum_even; otherwise add it to sum_odd.

e. Remove the last digit using temp = temp / 10.

f. Display Sum_even and Sum_odd.

8. For any other choice, display Invalid choice.

9. Stop.

Evaluation: The algorithm uses switch-case and digit extraction correctly. It produces the expected result for all prepared test cases.

Step 4: C Programming

```
#include <stdio.h>

int main()
{
   int n, choice, temp, digit;
   int even_count = 0, odd_count = 0;
   int sum_even = 0, sum_odd = 0;

   printf("Enter a positive integer: ");
   scanf("%d", &n);

   printf("1. Count even and odd digits\n");
   printf("2. Sum even and odd digits\n");
   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch (choice)
   {
      case 1:
         temp = n;
         while (temp > 0)
         {
            digit = temp % 10;
            if (digit % 2 == 0)
               even_count++;
            else
               odd_count++;
            temp = temp / 10;
         }
         printf("Even_dig = %d\n", even_count);
         printf("Odd_dig = %d\n", odd_count);
         break;

      case 2:
         temp = n;
         while (temp > 0)
         {
            digit = temp % 10;
            if (digit % 2 == 0)
               sum_even = sum_even + digit;
            else
               sum_odd = sum_odd + digit;
            temp = temp / 10;
         }
         printf("Sum_even = %d\n", sum_even);
         printf("Sum_odd = %d\n", sum_odd);
         break;

      default:
         printf("Invalid choice\n");
   }
   return 0;
```

```
}
```

Step 5: Compilation and Execution

The program was compiled and executed in Kali Linux and evaluated against the prepared test cases.

Step 6: Kali Linux Compilation and Execution

The following terminal-style screenshots show compilation and execution for the prepared test cases.

```
┌──(kali㉿kali)-[~]
└─$ nano day13.c
┌──(kali㉿kali)-[~]
└─$ gcc day13.c -o day13
┌──(kali㉿kali)-[~]
└─$ ./day13
Enter a positive integer: 12045
1. Count even and odd digits
2. Sum even and odd digits
Enter your choice: 1
Even_dig = 3
Odd_dig = 2
```

Test Case 2

```
┌──(kali㉿kali)-[~]
└─$ ./day13
Enter a positive integer: 12345
1. Count even and odd digits
2. Sum even and odd digits
Enter your choice: 1
Even_dig = 2
Odd_dig = 3
```

Test Case 3

```
┌──(kali㉿kali)-[~]
└─$ ./day13
Enter a positive integer: 12345
1. Count even and odd digits
2. Sum even and odd digits
Enter your choice: 2
Sum_even = 6
Sum_odd = 9
```